B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

Department of Computer Science & Engineering

(Data Science)

Bio-Inspired Algorithms(23DS7BSBIO) AAT Rubric (Total = 30 Marks)

Component	Criteria	Marks
1. Dataset Selection & Problem Definition (4 Marks)	Relevant dataset chosen & problem clearly defined	2
	Data preprocessing performed (cleaning, splitting, transformations)	2
2. ML/DL Model Development (6 Marks)	Suitable ML/DL model selected (CNN, ANN, SVM, RF, etc.)	2
	Correct architecture/model setup & justification	2
	Baseline results obtained without optimization (accuracy, RMSE, etc.)	2
3. Nature-Inspired Optimization Algorithm Implementation (10 Marks)	Correct choice of nature-inspired algorithm (GA/PSO/ACO/DE/BBO/SA/Firefly/etc.)	2
	Implementation of algorithm operators (selection, mutation, crossover / velocity update / pheromone update / migration, etc.)	4
	Optimization applied properly to tune model parameters (hyperparameters, weights, architecture parameters)	2
	Demonstrates improvement over baseline (accuracy, F1, loss, etc.)	2
4. Experimental Results, Analysis & Visualization (6 Marks)	Clear comparison: Baseline vs Optimized model	2
	Proper evaluation metrics used (Accuracy, Precision, F1, RMSE, AUC, etc.)	2
	Graphs/plots of convergence curves, learning curves, performance comparisons	2
5. Report Quality / Presentation / Code Documentation (4 Marks)	Well-organized report (introduction, methodology, results, conclusion)	2
	Clean, commented, modular code with reproducible results	2
Total		30 Marks